Mathematical analysis of the impact of the media coverage in mitigating the outbreak of COVID-19

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Séminaire ASA

Ouagadougou, 11/12/2022

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2. Model description
3. Mathematical study of the model
   - Stability results of the dfe
   - The endemic steady state
   - Stability results of the endemic equilibrium
4. Sensitivity analysis and herd immunity
5. Numerical simulation results
6. Conclusion

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- Appeared in November 2019 in Wuhan, mainland China, COVID-19 very quickly turned out to be a serious health problem around the world, with catastrophic consequences for the evolution of humankind.
- One of the major issues public health policy-makers meet in the implementation of any strategy for COVID-19 control is public adherence due to their perceived believes. To raise awareness in these uncertain times, one of the world's media priorities was to cover the pandemic.
- And even though lot of misinformation have circulated through social media and mass media, some of these informations led to behavioral changes and therefore made government measures more effective against the rapid spread of the virus (see N. Liu et al.).
- It is therefore evident that media coverage has had a greet influence in the disease dynamics. In light of that, we hereby consider the impact of the media in mitigating of the epidemic.

- In the literature, there several types of media function among which we have:

$$f(E, I, H) = e^{-\alpha_1 E - \alpha_2 I - \alpha_3 H} \tag{1}$$

proposed by Y. Liu et al. where α_1, α_2 and α_3, represent the effect of psychological impact of media reported numbers corresponding to exposed individuals E, infectious individuals I and hospitalized people H.

Yiping Liu and Ying-an Cui, The impact of media coverage on the dynamics of infectious diseases, *Int. J. Biomath.*, Vol. **1**, No. 1, (2007), pp. 65 – 74.

S. Samanta, S. Rana, A. Sharma, A. Misra, J. Chattopadhyay, Effect of awareness programs by media on the epidemic outbreaks: a mathematical model, *Appl. Math. Comput.*, Vol. **219**, No.12, (2013), pp. 965–6977.

- It was pointed out that the use of media can effectively reduce the contact rates among the population to a limited level, and the transmission rate

$$f(I) = \left(\beta - \frac{\beta_1 I}{m+I}\right) I \tag{2}$$

was developed by J. M. Tchuenche et al. to describe this inherent characteristics from media impact, where m indicates the velocity at which media and individuals respond to the epidemics, and β_1 is its reduced maximum value and less than β if I is sufficiently large.

J.M. Tchuenche, et al., The impact of media coverage on the transmission dynamics of human influenza, *BMC Public Health*, Vol. **11** No.1 (2011) S5.

- In this work, we introduce a new way of considering media impact on infectious diseases dynamics. The parameters $0 \leq M < 1$ accounting for media plays a similar role as vaccination

$$\lambda_v = (1 - M)\beta\frac{\eta_1 H + \eta_2 A + I}{N} \tag{3}$$

Obviously, this way of considering the media coverage within a model is one of the simplest way of doing it but it show some relevant aspects of media coverage impact on the disease global dynamics.

O. Koutou, A. B. Diabaté, B. Sangaré, Mathematical analysis of the impact of the media coverage in mitigating the outbreak of COVID-19, *Math. Comput. Simul.*, Vol. **205**, (2023), pp. 600 – 618.

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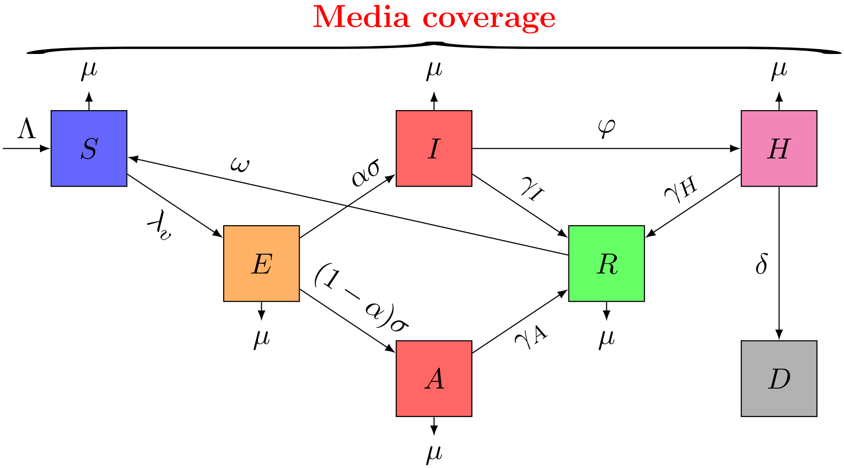
Model diagram



Model system

$$
\begin{cases}
\dot{S}(t) = \Lambda + \omega R - (1 - M)\beta \dfrac{\eta_1 H + \eta_2 A + I}{N} S - \mu S, \\
\dot{E}(t) = (1 - M)\beta \dfrac{\eta_1 H + \eta_2 A + I}{N} S - (\sigma + \mu)E, \\
\dot{I}(t) = \alpha\sigma E - (\varphi + \gamma_I + \mu)I, \\
\dot{A}(t) = (1 - \alpha)\sigma E - (\gamma_A + \mu)A, \\
\dot{H}(t) = \varphi I - (\delta + \gamma_H + \mu)H, \\
\dot{R}(t) = \gamma_I I + \gamma_A A + \gamma_H H - (\omega + \mu)R, \\
\dot{D}(t) = \delta H.
\end{cases}
\tag{4}
$$

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dfe and the basic reproduction number

The dfe of the model (4) is giving by $\mathcal{E}_0 = \left(\dfrac{\Lambda}{\mu}, 0, 0, 0, 0\right)$.

$$\mathcal{R}_0 = \mathcal{R}_I + \mathcal{R}_A + \mathcal{R}_H \tag{5}$$

$$\begin{aligned}
\mathcal{R}_I &= \frac{(1-M)\beta\alpha\sigma}{(\sigma+\mu)(\varphi+\gamma_I+\mu)},\\
\mathcal{R}_A &= \frac{(1-M)\beta\eta_2(1-\alpha)\sigma}{(\sigma+\mu)(\gamma_A+\mu)},\\
\mathcal{R}_H &= \frac{(1-M)\beta\eta_1\alpha\sigma\varphi}{(\sigma+\mu)(\varphi+\gamma_I+\mu)(\delta+\gamma_H+\mu)}.
\end{aligned}$$

P. van den Driessche & J. Watmough, Reproduction numbers and sub-threshold endemic equilibria for compartmental models of disease transmission, *Math. Biosci.*, Vol. **180**, (2002), pp. 29 − 48.

Stability results of the dfe

Theorem

If $\mathcal{R}_0 < 1$ then the disease-free equilibrium $\mathcal{E}_0$ is locally asymptotically stable and unstable if $\mathcal{R}_0 > 1$.

Theorem

If $\mathcal{R}_0 < 1$ then the no-disease equilibrium point $\mathcal{E}_0$ is globally asymptotically stable.

Proof:Using the following Lyapunov candidate

$$\begin{aligned} L &= E + \left(\frac{(1-M)\beta}{\varphi+\gamma_I+\mu} + \frac{(1-M)\beta\eta_1\varphi}{(\varphi+\gamma_I+\mu)(\delta+\gamma_H+\mu)} \right) I \\ &+ \frac{(1-M)\beta\eta_2}{\gamma_A+\mu} A + \frac{(1-M)\beta\eta_1}{\delta+\gamma_H+\mu} H. \end{aligned}$$

and noticing that $\dfrac{S}{N} \leq 1$, the above result holds.

The endemic equilibrium

Theorem

System (4) admits a unique positive endemic equilibrium $\mathcal{E}^* = (S^*, E^*, I^*, A^*, H^*, R^*)$ *whenever* $\mathcal{R}_0 > 1$.

Proof By putting the right hand side of system (4) to zero, and keeping each state variable different from zero ($S \neq 0$, $E \neq 0$, $I \neq 0$, $A \neq 0$, $H \neq 0$ and $R \neq 0$) then one obtains

$$S^* = \frac{b_4\Lambda - \delta\varphi I^*}{\mu b_4 \mathcal{R}_0}, \; E^* = \frac{b_2}{\alpha\sigma} I^*, \; A^* = \frac{(1-\alpha)b_2}{\alpha b_3} I^*, \; H^* = \frac{\varphi}{b_4} I^*$$

$$R^* = \frac{\alpha\gamma_I b_3 b_4 + (1-\alpha)\gamma_A b_2 b_4 + \alpha\varphi\gamma_H b_3}{\alpha(\omega+\mu)b_3 b_4} I^*$$

and

$$I^* = \frac{\alpha(\omega+\mu)b_3 b_4 \Lambda(\mathcal{R}_0 - 1)}{\alpha b_3 b_4 \mathcal{Q}_1 + (1-\alpha)b_2 b_4 \mathcal{Q}_2 + \alpha\varphi b_3 \mathcal{Q}_3},$$

The endemic equilibrium point

with

$$\begin{aligned}\mathcal{Q}_1 &= (1-M)\beta(\omega+\mu) - \omega\gamma_I\mathcal{R}_0,\\ \mathcal{Q}_2 &= (1-M)\beta\eta_2(\omega+\mu) - \omega\gamma_A\mathcal{R}_0,\\ \mathcal{Q}_3 &= \left((1-M)\beta\eta_1 - \delta\right)(\omega+\mu) - \omega\gamma_H\mathcal{R}_0.\end{aligned}$$

It is obvious that $\mathcal{R}_0 = 1$ leads to the disease-free equilibrium, whereas when $\mathcal{R}_0 > 1$ then there exists a unique endemic equilibrium.

S. E. Eikenberry, M. Mancuso, E. Iboi, T. Phan, K. Eikenberry, Y. Kuang, E. Kostelich, A. B. Gumel, To mask or not to mask: Modeling the potential for face mask use by the general public to curtail the COVID-19 pandemic, *Infectious Disease Modelling*, Vol. **5**, (2020), pp. 293 – 308.

Stability results of the endemic equilibrium point

Theorem

If $\mathcal{R}_0 > 1$, then the endemic equilibrium point $\mathcal{E}^*$ is locally asymptotically stable.

Proof: The Routh-Hurwitz criterion.

Theorem

Let $\mathcal{R}_0 > 1$. Whenever $sign(N - N^*) = sign(H - H^*)$ then the endemic equilibrium point $\mathcal{E}^*$ of system (4) is globally asymptotically stable.

Proof: We use the Lyapunov's functional method.

Stability results of the endemic equilibrium point

Proof:For that, we consider

$$\mathcal{V} = \sum_{i=1}^{6} (x_i - x_i^*) - \sum_{i=1}^{6} x_i^* \ln \left(\sum_{i=1}^{6} x_i / \sum_{i=1}^{6} x_i^* \right) \qquad (6)$$

where $x_i, \; i = 1, 2, ..., 6$ denote the state variables.
One can easily check that the above function is null at the endemic equilibrium and positive elsewhere. Moreover, it can also be rewritten as follows:

$$\mathcal{V} = N - N^* - N^* \ln \frac{N}{N^*}.$$

Cruz Vargas-De-León, Volterra-type Lyapunov functions for fractional-order epidemic systems, *Commun. Nonlinear Sci. Numer. Simulat.*, Vol. **24**, (2015), pp. 75 – 85.

Stability results of the endemic equilibrium point

Therefore, the derivative form of the function $\mathcal{V}$ is given as

$$
\begin{aligned}
\dot{\mathcal{V}} &= \left(1 - \frac{N^*}{N}\right)\dot{N} \\
&= \left(1 - \frac{N^*}{N}\right)(\Lambda - \mu N - \delta H) \\
&= \frac{N - N^*}{N}(\mu N^* + \delta H^* - \mu N - \delta H) \\
&= \frac{N - N^*}{N}[-\mu(N - N^*) - \delta(H - H^*)] \\
&= -\frac{\mu(N - N^*)^2}{N} - \frac{\delta(N - N^*)(H - H^*)}{N} \\
&\leq 0.
\end{aligned}
$$

Consequently, the solution $\mathcal{E}^*$ is globally asymptomatically stable

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Sensitivity analysis

Performing a sensitivity analysis helps us to understand how changes in the model parameters will affect the disease spreading dynamics. And to carry it out, a simple approach is used to calculate the sensitivity index of each parameter that appeared in $\mathcal{R}_0$ through out the following formula:

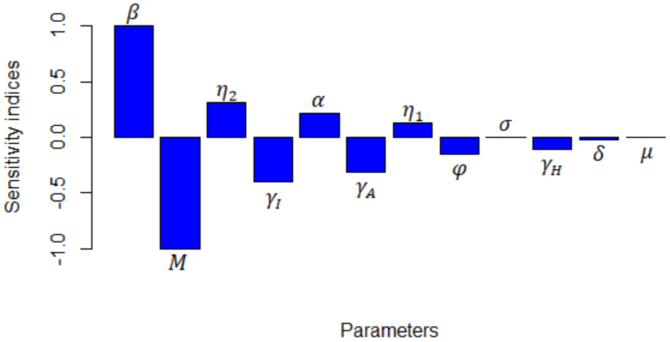
$$\chi_p^{\mathcal{R}_0} = \frac{\partial \mathcal{R}_0}{\partial p} \times \frac{p}{\mathcal{R}_0} \simeq \frac{\%\Delta\mathcal{R}_0}{\%\Delta p}. \tag{7}$$

J. K. K. Asamoah, M.A. Owusu, Z. Jin, F.T. Oduro, A. Abidemi, E.O. Gyasi, Global stability and cost-effectiveness analysis of COVID-19 considering the impact of the environment: using data from Ghana, *Chaos Solitons Fractals*, Vol. **140**, (110103) (2020).

Sensitivity analysis

Param.	Val.	Ref.	Dimen.	Sens. indices
Λ	200	estimated	humans/day	-
ω	0.42	estimated	/day	-
β	0.55	[2]	/day	+1
η_1	0.2	[2, 3]	-	+0.1327
η_2	0.5	[3]	-	+0.3137
σ	1/6	[2]	/day	+0.00025
α	0.6	[3]	-	+0.2158
φ	0.1	[3, 2]	/day	-0.1499
γ_A	1/7	[3, 2]	/day	-0.3136
γ_I	1/7	[3, 2]	/day	-0.4036
γ_H	1/14	[3, 2]	/day	-0.1135
μ	4.25×10^{-5}	estimated	/day	-0.00055
δ	0.012	[2]	/day	-0.0191
M	50%	estimated	-	-1

Global sensitivity indices diagram for $\mathcal{R}_0$



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Herd immunity threshold

The course of a communicable disease may be slowed down or the disease may somewhat be blocked from being spread within an area when a certain rate of people is being recovered from the disease with a permanent immunity. This phenomenon called group immunity or herd immunity represents some kind of indirect protection for susceptible people due to their scarcity.
Herd immunity threshold is related to the basic reproduction number and it can be computed as follows

$$\frac{\mathcal{R}_0 - 1}{\mathcal{R}_0} = 1 - \frac{1}{\mathcal{R}_0}.$$

Therefore, the proportion of the population that need to be infected to reach the group immunity level is $(1 - 1/\mathcal{R}_0)\,\%$. In this study if there is no media coverage $\mathcal{R}_0 \simeq 2.45$; so herd immunity normally occurs when about 59% of the total population is infected.

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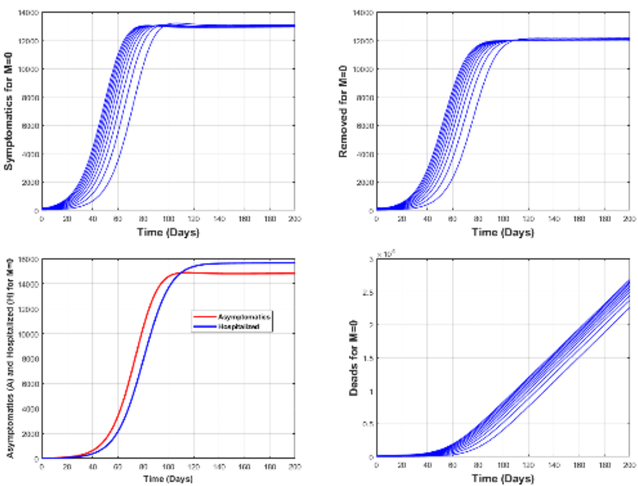

Figure: Global dynamics of the disease when the media coverage has no effect on people behaviours at all. For this first case, $\mathcal{R}_0 \simeq 2.45$.

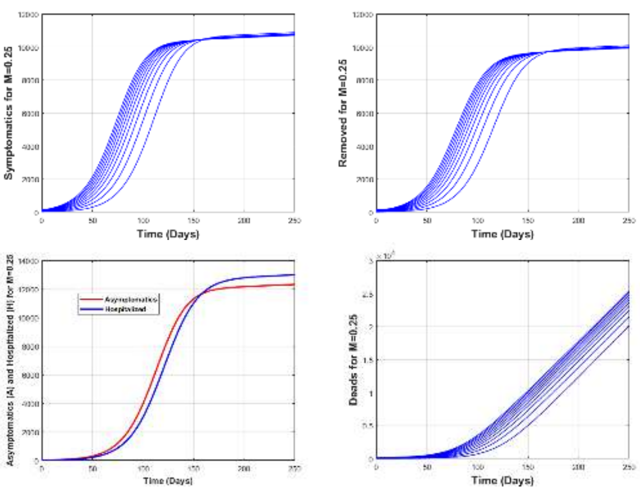

Figure: Global dynamics of the disease when the media coverage efficiency rate is about 25%. We therefore have $\mathcal{R}_0 \simeq 1.84$.

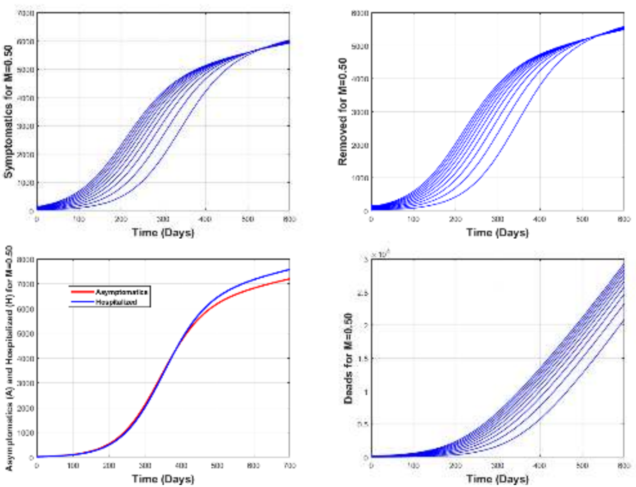

Figure: Global dynamics of the disease when the efficiency rate of media coverage is up to 50%. In this case, we obtain $\mathcal{R}_0 \simeq 1.22$.

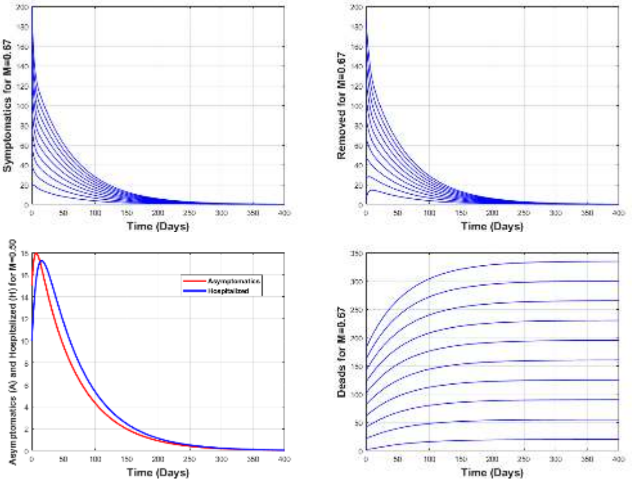

Figure: Global dynamics of the disease when the rate of efficiency of the media coverage is up to 67%. So, $\mathcal{R}_0 \simeq 0.81$.

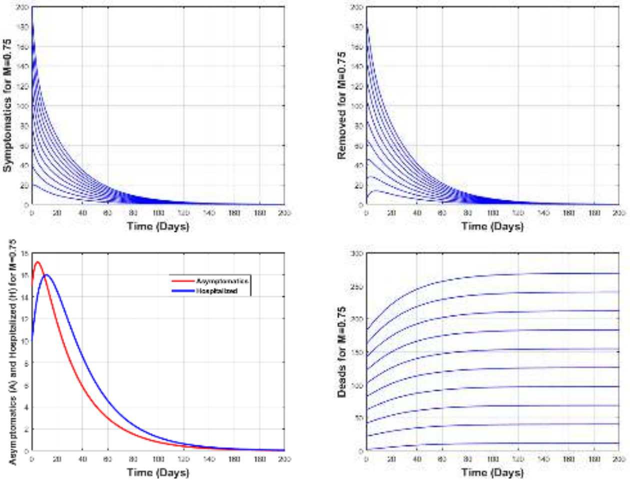

Figure: Global dynamics of the disease when the media coverage efficiency rate reaches $M = 75\%$, such that $\mathcal{R}_0 \simeq 0.61$.

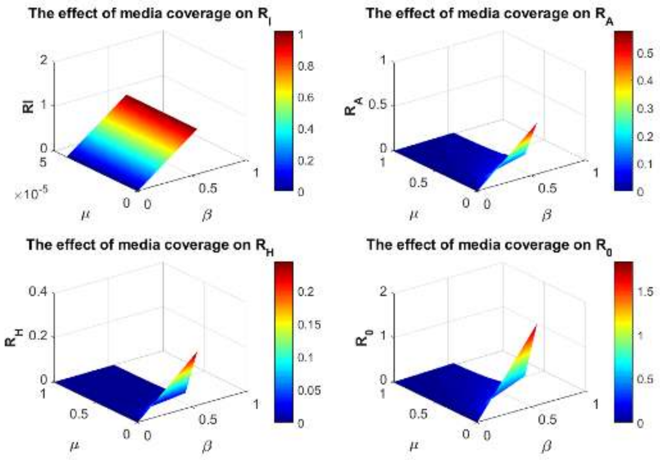

Figure: Evolution of the threshold parameters when the efficiency rate of the media coverage is $M = 25\%$.

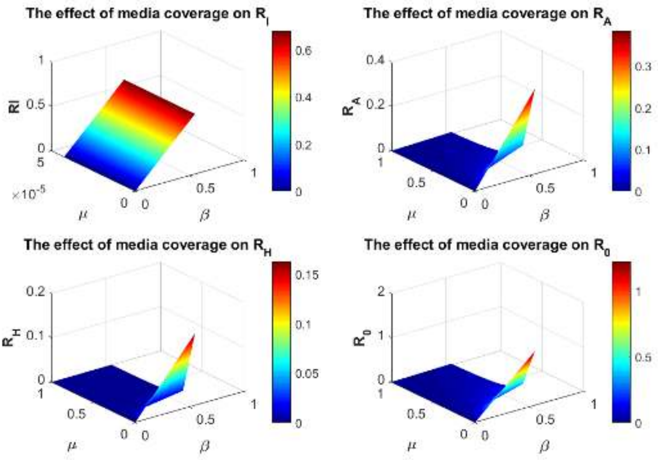

Figure: Evolution of the threshold parameters when the efficiency rate of the media coverage is $M = 50\%$.

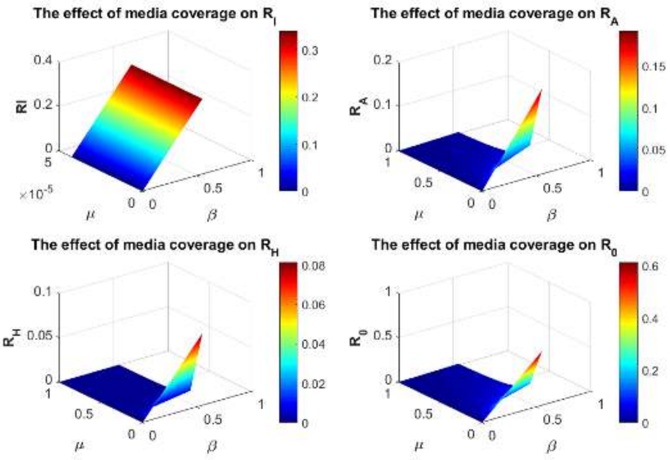

Figure: Evolution of the threshold parameters when the efficiency rate of the media coverage is $M = 75\%$.

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Conclusion

- How media coverage does affect the course of the current epidemic of the COVID-19.
- Global sensitivity analysis and herd immunity threshold computation
- We used different scenarios for the model simulation and the results have shown that being aware and respecting safety guidelines such as wearing of masks, social distancing, frequent disinfection of hands with water and soap or using any other means to keep from being infected and reducing as much as possible human mobility has positive effect in the combat against the coronavirus propagation

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